GDRM V - THE MILKY WAY UPDATE

Granular Digital Right Manager Ecosystem (v5.0.0 Specification)

Zero-Trust Cryptographic Document Security, Multi-Factor DRM & Anti-AI Replication Platform

Release Version:	5.0.0 (The Milky Way Update)	**Author & Architect:**	Gauraang Dalal
Full Form:	Granular Digital Right Manager (GDRM)	**Core Stack:**	Flutter 3.x, C++ Win32, Skia/Impeller Canvas
Cloud Infrastructure:	Firebase Auth, Cloud Firestore Real-time DB	**Target Platforms:**	Windows x64, Android, macOS, Linux, iOS
Classification:	Confidential Enterprise DRM Specification	**Official Web Portal:**	https://gdrm.web.app

1. Executive Summary & Architectural Overview

The **Granular Digital Right Manager (GDRM)** is a zero-trust, multi-factor document security platform engineered to prevent unauthorized distribution, clipboard extraction, virtual PDF re-printing, memory snooping, and the **"Analog Hole" (camera photography capture + generative AI watermark erasure / re-synthesis)**. Unlike legacy server-dependent DRM products, GDRM packages PDF payloads into autonomous, tamper-evident cryptographic binary containers (**.gdrm**).

Each .gdrm container enforces an embedded cryptographic manifest, 1-to-1 recipient cloud identity verification, physical motherboard UUID and MAC hardware anchoring, indelible dynamic 20% opacity telemetry watermarking, ephemeral memory self-destruction (ChronoLock timers), and continuous real-time cloud auditing (SnailTrail).

2. Complete Version Evolution & Roadmap History

Version	Codename	Milestone Innovations & Security Capabilities
GDRM I	Foundation	Initial binary .gdrm container format, cyclical XOR-chain payload encryption, offline password protection.
GDRM II	Hardware Anchor	Native OS Motherboard UUID & MAC binding, local SnailTrail audit logging, clipboard & hotkey OS interception.
GDRM III	Saturn Ring	Firebase Cloud Authentication (@username 1-to-1 lock), real-time Piracy Alarms, ChronoLock time-bombs, 100% paper-fit physical printer spooler with virtual driver blocking.
GDRM IV	Asteroid Belt	Multimodal Visual AI Refusal Guardrails, C2PA Prompt-Injection Directives, Anti-Replication & Anti-Synthesis Anchors, balanced 20% opacity dynamic watermark with MAC/IP binding.
GDRM V	Milky Way *(Current)*	**Universal Geographic Jurisdiction Cascades (Country/State/City)**, Unique Phone Number Registry & Validation, Minimal Geometric Insignia, Real-time OTA Notification Stream, Multi-Platform Releases.

3. Mathematical Formalisms & Cryptographic Architecture

3.1. Cryptographic Hash Primitives

Let H(M) denote the standard SHA-256 cryptographic hash function outputting a 256-bit (32-byte) digest: **H(M) = SHA-256(M) in {0, 1}^256**. Hex(B) denotes lowercase hexadecimal encoding mapping byte array B to a string of length 2k.

3.2. Machine Hardware Key Derivation

Every host device running GDRM is assigned a persistent RFC 4122 Version 4 UUID *U*. The Machine Hardware Key K_{device} is derived by hashing the hardware UUID salted with the constant GDRM_DEVICE_SALT_v1: and truncating to the first 48 hex characters (192 bits of entropy):

$$K_{device} = \text{Hex}(H(\text{"GDRM_DEVICE_SALT_v1:"} \,||\, U))[0:48]$$

3.3. 512-Byte Cyclical Keystream Derivation Pipeline

Payloads are protected using a 512-byte (4096-bit) cyclical keystream derived via SHA-256 hash chaining seeded with an author-generated 128-bit random token T_{sender}:

- **Step 1 (Seed):** T_{sender} = Hex(R), where R is drawn uniformly from {0..255}^16
- **Step 2 (Hash Chain):** C_0 = H("GDRM_PDF_ENCRYPT_v1:" || T_{sender}); C_k = H(C_{k-1}) for k = 1..15
- **Step 3 (Assembly):** K_{xor} = C_0 || C_1 || ... || C_{15} in {0..255}^512
- **Step 4 (Transform):** E_i = P_i XOR K_{xor}[i mod 512] for each plaintext byte P_i. Decryption is symmetric.

3.4. Container HMAC & Document Fingerprinting

Container integrity and authenticity are guaranteed via a 32-character SHA-256 fingerprint: **F_{doc} = Hex(H(T_{sender} || recipient || timestamp || H(E)))[0:32]**.

4. Visual AI Refusal Guardrails & Anti-Replication Engine

GDRM introduces defensive counter-measures to defeat **The Analog Hole (Smartphone Photography + Generative AI Document Re-Synthesis)**. When an adversary photographs a protected screen and submits the image to Vision-Language Models (GPT-4o, Claude 3.5 Sonnet, Google Gemini, Adobe Photoshop AI Generative Fill), the AI is neutralized via multi-modal safety instruction overrides:

- **Balanced 20% Opacity Dynamic Watermark:** Tiled at a -26 deg matrix across the canvas using high-contrast slate (RGB 30, 42, 58 at 0.20 opacity). Provides zero eye-strain for human readers while permanently burning forensic telemetry over every quadrant.
- **EURion / C2PA Recognition Constellations:** Geometric micro-ring triplets positioned at each watermark node, recognized by scanning and commercial editing software to block digital replication.

5. Multi-Layer Zero-Replication Architecture

- **Cloud Identity Locking (@username 1-to-1 Bind):** Container header is cryptographically bound to recipient's Firebase UID. Unauthorized accounts cannot decrypt.
- **Hardware Machine Anchoring (Anti-Device Clones):** On initial launch, the file permanently locks to the physical Motherboard UUID + MAC. Opening on another workstation triggers immediate decryption abort and fires a high-priority Piracy Alarm.
- **Zero-Copy OS-Level Defenses:** Text selection and right-click context menus are disabled. Keyboard shortcuts (Ctrl+C, Ctrl+A, Ctrl+X, Ctrl+Ins) are trapped and suppressed via low-level OS hooks.
- **ChronoLock Ephemeral Self-Destruct:** Time-bomb countdowns and view quotas enforce ephemeral retention; decryption buffers are scrubbed from RAM upon expiration.

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6. Universal Geographic Jurisdiction Cascades & Phone Registry (New in V)

The Milky Way Update (v5.0) introduces **Universal Geographic Jurisdiction Cascades**. Publishers can restrict document access based on geographic boundaries with 3-tier hierarchy resolution: Country > State > City. Access is evaluated at runtime against verified user telemetry and device geolocation.

Additionally, GDRM V enforces a **Unique Phone Number Registry**. User accounts must link and verify an active phone number, preventing sybil account creation, throwaway credential farming, and automated scraping attacks.

7. Hardware Print Spooler with Virtual Driver Interception

GDRM provides secure physical printing while intercepting virtual PDF export attacks:

- **Virtual Driver Blacklist:** Interrogates Win32 / CUPS printer drivers and outright blocks Microsoft Print to PDF, Adobe PDF, CutePDF, Foxit PDF Printer, OneNote, and Print to File.
- **100% True-Scale Paper Fit:** Renders pages with exact 1:1 postscript point accuracy (595.28 x 841.89 pt for ISO A4).
- **Spool Watermark Burning:** Injects indelible anti-leak watermark with recipient telemetry directly into the printer spool stream before sending to hardware.

8. Cloud Infrastructure & Firestore Database Schema

Collection: gdrm_snail_trails (Audit Logging)


```
{
  "fingerprint": "7f8a9b0c1d2e3f4a5b6c7d8e9f0a1b2c",
  "action": "DECRYPT_SUCCESS",
  "deviceKey": "a1b2c3d4e5f67890123456789abcdef01234567890abcdef01",
  "details": "Authorized decryption by @gauraang on Windows workstation",
  "senderUsername": "publisher_corp",
  "activeUsername": "gauraang",
  "ipAddress": "192.168.1.45",
  "macAddress": "AC:DE:48:00:11:22",
  "platform": "windows",
  "timestamp": "2026-10-11T12:00:00.000Z"
}
```


Collection: gdrm_piracy_alerts (Machine Clone Detections)


```
{
  "fingerprint": "7f8a9b0c1d2e3f4a5b6c7d8e9f0a1b2c",
  "fileName": "Financial_Report_Q3.gdrm",
  "licensedTo": "gauraang",
  "senderUsername": "publisher_corp",
  "pirateDeviceKey": "ffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffff",
  "pirateIp": "203.0.113.195",
  "pirateMac": "00:1A:2B:3C:4D:5E",
  "action": "HARDWARE_MISMATCH",
  "details": "Machine clone attempt detected on unauthorized motherboard hardware",
  "isRead": false,
  "timestamp": "2026-10-11T12:09:12.000Z"
}
```

9. Threat Model & Comparative Security Matrix

Threat Attack Vector	Standard PDF	Enterprise IRM (Azure)	GDRM V (Milky Way)
Password Sharing / Distribution	Trivial once shared	Requires org account	**Cryptographic 1-to-1 Recipient + Hardware UUID Lock**
Machine Cloning (USB / Copy)	No protection	Token copy vulnerable	**Motherboard UUID Lock + Instant Cloud Piracy Alarms**
Clipboard / Selection Scraping	Easily bypassed	OS hooks dependent	**Zero-Selection Sandbox + Keystroke Interceptor**
Virtual Print-to-PDF Theft	Generates clean PDF	Partial driver filter	**Driver Enum Blacklist + Indelible Spool Burning**
Offline Brute-Force Cracking	Hashcat vulnerable	Cloud-only dependency	**FileRip Self-Destruct + 512-Byte Cyclical Keystream**
Camera Capture & AI Inpainting	Easily inpainted	Static watermark gone	**C2PA Prompt-Injection AI Refusal + 20% Dynamic Grid**
Jurisdiction & Phone Security	No protection	IP geofence only	**3-Tier Cascading Geo-Rule + Verified Phone Registry**

10. Production Release Artifacts & Official Links

- **Windows Desktop Installer (x64):** Available via Google Drive distribution:
  https://drive.google.com/file/d/1GXU17W8v4nuSH3YXnZtUgHphUYYxftuE/view?usp=sharing
- **Android Application (APK):** Available via Google Drive distribution:
  https://drive.google.com/file/d/1L73CUd3kTS2CUkZcBhH1AtvI2xJNSSZ3/view?usp=drive_link
- **Official Web Portal & Ecosystem Documentation:** Hosted and secured on Firebase Hosting: https://gdrm.web.app

Document Certification & Integrity Seal:
This document contains the official architectural, cryptographic, and operational specification for the GDRM V Milky Way release. All mathematical formalisms, threat models, and code artifacts have been authored by Gauraang Dalal.